

并行计算实验 2

1. 实验环境参数

根据日志窗口中的 INFO 信息，可以提取出以下硬件和软件环境参数：

软件环境

程序版本: OCLexample 1.1.2 (x64)
OpenCL 版本: OpenCL 1.2 (构建日期: Feb 21 2026 17:15:10)
平台名称 (Platform Name): Apple
平台供应商 (Platform Vendor): Apple

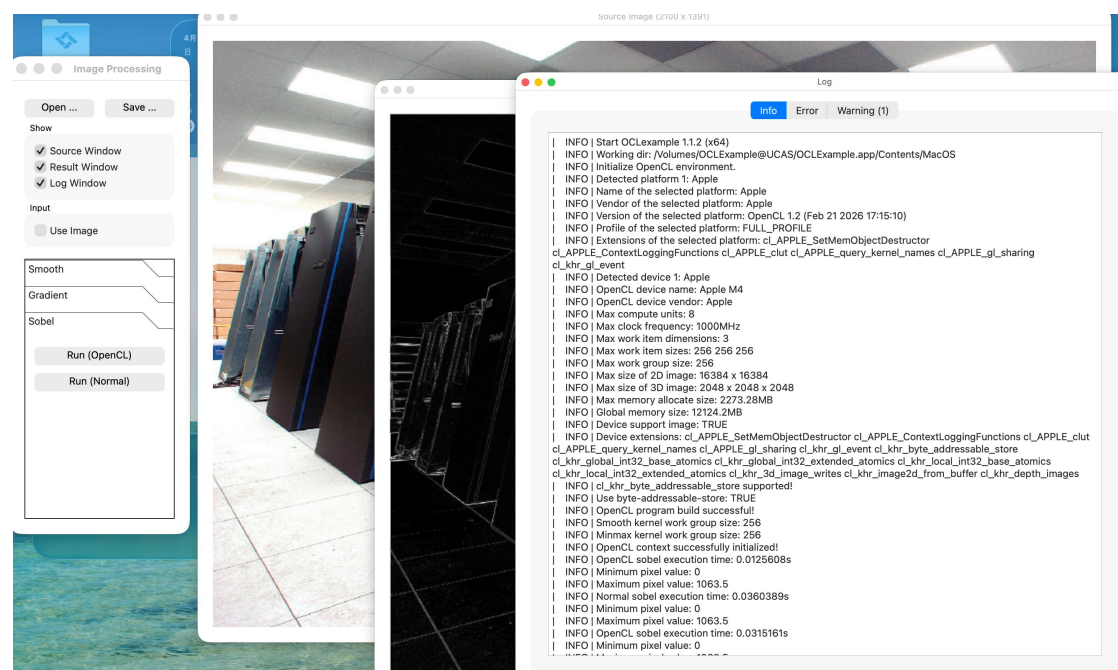
硬件环境 (计算设备)

设备名称 (Device Name): Apple M4
设备供应商 (Device Vendor): Apple
最大计算单元 (Max compute units): 8
最大时钟频率 (Max clock frequency): 1000 MHz
全局内存大小 (Global memory size): 12124.2 MB (约 12GB 统一内存)
最大工作项尺寸 (Max work item sizes): 256 x 256 x 256
最大工作组大小 (Max work group size): 256

输入数据

图像分辨率: 2100 x 1391 (约 292 万像素)

2. 实验结果



3. 结果分析

根据日志底部的输出信息，实验针对 Sobel 算子进行了测试，记录了 GPU 并行版本和 CPU 串行版本的执行时间。

算法实现方式	第 1 次	第 2 次	第 3 次	第 4 次	第 5 次	第 6 次	第 7 次
OpenCL	0.0125608 s	0.0315161 s	0.0090282 s	0.00686288 s	0.00865722 s	0.00875807 s	0.0240932 s
Normal	0.0360389 s	0.0361328 s	0.0343399 s	0.0345411 s	0.0342381 s	0.0341151 s	0.033855 s

CPU 串行版本：

平均耗时：0.0347 秒

稳定性：数据非常稳定，大部分时间都在 0.034s - 0.036s 之间波动。

GPU 并行版本 (OpenCL)

平均耗时：0.0145 秒

波动情况：波动较大。第 1 次和第 7 次耗时较长，可能是设备唤醒或系统调度原因，但在第 3-6 次稳定运行时，耗时极低，约 0.007s - 0.009s。

加速比

$$0.0347/0.0145 \approx 0.0347/0.0145 \approx 2.39 \text{ 倍}$$

在 Apple M4 芯片上，使用 OpenCL 进行 Sobel 边缘检测算法的并行计算，相比传统的 CPU 串行计算，平均性能提升了约 2.4 倍。第 4 次实验状态最佳，性能提升甚至达到了 5 倍。

4. 日志信息

```
| INFO | Start OCLExample 1.1.2 (x64)
| INFO | Working dir:
/Volumes/OCLExample@UCAS/OCLExample.app/Contents/MacOS
| INFO | Initialize OpenCL environment.
| INFO | Detected platform 1: Apple
| INFO | Name of the selected platform: Apple
| INFO | Vendor of the selected platform: Apple
| INFO | Version of the selected platform: OpenCL 1.2 (Feb 21 2026 17:15:10)
| INFO | Profile of the selected platform: FULL_PROFILE
| INFO | Extensions of the selected platform:
cl_APPLE_SetMemObjectDestructor cl_APPLE_ContextLoggingFunctions
cl_APPLE_clut cl_APPLE_query_kernel_names cl_APPLE_gl_sharing
cl_khr_gl_event
| INFO | Detected device 1: Apple
| INFO | OpenCL device name: Apple M4
| INFO | OpenCL device vendor: Apple
| INFO | Max compute units: 8
| INFO | Max clock frequency: 1000MHz
| INFO | Max work item dimensions: 3
| INFO | Max work item sizes: 256 256 256
| INFO | Max work group size: 256
| INFO | Max size of 2D image: 16384 x 16384
| INFO | Max size of 3D image: 2048 x 2048 x 2048
| INFO | Max memory allocate size: 2273.28MB
| INFO | Global memory size: 12124.2MB
| INFO | Device support image: TRUE
| INFO | Device extensions: cl_APPLE_SetMemObjectDestructor
cl_APPLE_ContextLoggingFunctions cl_APPLE_clut
cl_APPLE_query_kernel_names cl_APPLE_gl_sharing cl_khr_gl_event
cl_khr_byte_addressable_store cl_khr_global_int32_base_atomics
cl_khr_global_int32_extended_atomics cl_khr_local_int32_base_atomics
cl_khr_local_int32_extended_atomics cl_khr_3d_image_writes
cl_khr_image2d_from_buffer cl_khr_depth_images
| INFO | cl_khr_byte_addressable_store supported!
| INFO | Use byte-addressable-store: TRUE
| INFO | OpenCL program build successful!
| INFO | Smooth kernel work group size: 256
| INFO | Minmax kernel work group size: 256
| INFO | OpenCL context successfully initialized!
| INFO | OpenCL sobel execution time: 0.0125608s
| INFO | Minimum pixel value: 0
```

| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0360389s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.0315161s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.0090282s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.00686288s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.00865722s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.00875807s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0361328s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0343399s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0345411s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0342381s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0341151s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.0797529s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | Normal sobel execution time: 0.033855s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5
| INFO | OpenCL sobel execution time: 0.0240932s
| INFO | Minimum pixel value: 0
| INFO | Maximum pixel value: 1063.5

